ATLAS INSTITUTIONAL CRYPTO SCREENER

SYSTEM ARCHITECTURE & QUANTITATIVE STRATEGY SPECIFICATION (V2.6)

Status: Production Institutional Release
Venue: Binance USD-M & Spot Universes
Execution: Two-Speed Algorithmic Radar (21.6s Scan)

1. EXECUTIVE SUMMARY & THE QUANTITATIVE EDGE

The Atlas Institutional Crypto Screener is a high-throughput, multi-timeframe quantitative screening engine engineered to systematically discover high-expectancy setups across Binance USD-M perpetuals. Rather than chasing post-expansion volatility (+20% to +50% pumps) where retail traders become exit liquidity, Atlas detects:

- **Long Opportunities:** Pre-breakout volatility coils, Wyckoff Phase C springs, and quiet volume-profile absorption *before* trend markup.
- **Short Opportunities:** Vertical run-up blow-off tops, Wyckoff Phase C Upthrusts After Distribution (UTAD), and multi-timeframe exhaustion in the upper 55% to 100% of the 30-day range.

2. CLEAN HEXAGONAL ARCHITECTURE & TWO-SPEED ENGINE

Atlas implements strict Hexagonal separation (Ports & Adapters) with a **Two-Speed Screening Model**:

1. Domain Layer
Pure mathematical models: `CryptoOpportunity`, `Direction`, `SetupLabel`, `Candle`, `IndicatorSnapshot`, and `ChecklistVerification`. Completely zero external dependencies.

2. Application Core
`CryptoOpportunityScreener` coordinates 2-stage filtering: Stage 1 evaluates 1H/4H price action without network latency; Stage 2 enriches top candidates with full derivatives.

3. Two-Speed Radar
Core On-Demand: Sweeps top liquid universe in **21.6 seconds**.
Full Cron Sweep: Runs automatically on aligned 4-Hour UTC candle closes.

3. UNIVERSE SELECTION & LIQUIDITY FILTERING FUNNEL

From an unfiltered raw universe of 740+ instruments on Binance, Atlas applies rigorous multi-stage hygiene gates before computing quantitative factors:

Funnel Stage	Filtering Mechanism	Institutional Rationale & Live Count
1. Total Universe	Active Binance USD-M perpetual futures.	**742 instruments** tracked across the live futures exchange.
2. Hygiene & Asset Gate	Excludes gold (`PAXG`), synthetic commodities (`XAU` , `COPPER`), benchmark pairs (`BTC` , `ETH`), and fiat/stables.	**553 eligible contracts** remaining; eliminates non-crypto asset volatility and synthetic drift.
3. Liquidity & Spread Floor	Min 24h quote volume of **$500,000**; max bid-ask spread capped at **50 bps**.	**444 liquid contracts**; eliminates illiquid wash pairs vulnerable to extreme stop-loss slippage.
4. Canonical Deduplication	Best quote volume selection per underlying asset (retaining best of USDT vs USDC).	Eliminates redundant multi-quote scanning and concentrates liquidity on the primary contract.
5. Deep Evaluation Universe	Ranked by **Institutional Liquidity Index (ILI)** and active expansion flags.	Top **250 liquid contracts** evaluated concurrently across 1H, 4H, and 30-day Volume Profiles.

4. MULTI-TIMEFRAME TELEMETRY & VECTORIZED PROCESSING

For every selected asset, Atlas ingests 1H and 4H candle series concurrently, deriving 1D and 1W structures in-memory:

Vectorized High-Performance Ingestion: 1H Indicator Snapshot (EMA9/20/50/200, ATR14, MACD, ADX, DMI, Normalized Momentum) • Inlined 4H 180-Bar VPVR (POC, VAH, VAL calculated in <66ms per token) • Resampled 1D/1W Macro Context • 7-Day & 1-Hour Open Interest Delta • Taker Buy/Sell Aggression.

QUANTITATIVE LONG SCREENING STRATEGY

PRE-BREAKOUT VOLATILITY SQUEEZE & WYCKOFF PHASE C ACCUMULATION

Direction: LONG Setups
Horizon: Multi-Day Swing (1–5 Days)
Min Conviction: 65.0 Floor (Grade B+)

1. THE PRE-BREAKOUT PHILOSOPHY

Traditional retail screeners alert on tokens *after* they appear on "Top Gainers" (+20% to +40% daily pumps). By that point, smart money is actively distributing inventory into breakout chasers. The Atlas Long engine operates on the opposite principle: identify **volatility compression**, **supply exhaustion**, and **order-flow absorption** *before* the volatility expansion occurs.

2. THE 6 MANDATORY QUANTITATIVE LONG GATES

Gate Name	Mathematical Formulation	Institutional Function
Gate 1: Minervini Trend Baseline	`Close >= 4H_EMA50 * 0.985` `4H_EMA20 >= 4H_EMA50 * 0.975`	Mandates Stage 1 accumulation or Stage 2 markup. Eliminates Stage 4 downtrends and falling knives.

Gate 2: Anti-Chasing 2.0	`Excursion_72h <= 36%`
	`Ret_24h in [-8%, +14%]`
	`RSI_1H <= 76 & RSI_4H <= 70`
		Disqualifies parabolic blow-off pumps and post-pump exhaustion wicks. Ensures price is tightly coiled.

Gate 3: Base Duration & Ceiling Proximity	`Range_Height_30d <= 65%` `Dist_to_Ceiling in [-3%, +35%]`	Verifies multi-week consolidation base. Price must be resting beneath resistance, ready for expansion.
Gate 4: Value Area Acceptance	`Close >= 30D_POC * 0.975` `Value_Area_Pos >= 25%`	Confirms price is accepted above the 30-day Point of Control. Rejects tokens trading beneath heavy overhead volume.
Gate 5: Volatility Compression	`BB_Inside_KC == True` or `Bandwidth < 28% (Ideal < 10%)`	Identifies John Carter Squeeze geometry: Bollinger Bands contracting completely inside Keltner Channels.
Gate 6: S/R Breakout Resistance Gate	`if Dist_to_Ceiling <= 4.5%:` `  OI_7d >= -1% & RVOL >= 1.15x`	**Critical Bull-Trap Filter:** If price tests ceiling resistance while OI is bleeding and volume is flat, it is rejected immediately. Breakouts require expanding OI.

3. LONG SETUP ARCHETYPES

`COIL_SPRING_LONG` **Wyckoff Phase C**

Price conducts a terminal shakeout/spring test below support, rapidly reclaims the 4H EMA50 / POC shelf, with down-volume drying up (VCR ≤ 0.90x).

`BREAKOUT_LONG` **Volume Expansion**

Price breaks above local swing pivot / VAH ceiling with relative volume expansion (≥ 1.25x), positive normalized momentum, and expanding open interest.

`MOMENTUM_EMERGING_LONG` **Residual Alpha**

Early phase regime shift with strong positive BTC-residual alpha (≥ +3% over 24h), bullish MACD histogram expansion, and aggressive taker buying (>1.15x).

`OI_ACCUMULATION_LONG` **Smart Money Absorption**

Continuous institutional accumulation characterized by massive 7-day open interest expansion (+10% to +60%) while price remains compressed within a tight base.

4. EXECUTION BRACKET CONSTRUCTION (ASYMMETRIC R:R >= 2.0 FLOOR)

Every qualifying long candidate is stamped with an institutional execution bracket:

- **Retest Entry (Limit):** Anchored at the 4H EMA50 / POC support shelf (`Primary_Shelf + 0.15 * ATR`).
- **Breakout Trigger (Stop-Buy):** Placed immediately above local compression pivot resistance (`Resistance + 0.05 * ATR`).
- **Invalidation Stop Loss:** Placed structurally below the recent 4H swing low shelf (`Support - 0.35 * ATR`).
- **Multi-Tier Targets: Target 1:** 30d High / VAH ceiling • **Target 2:** Fibonacci 1.618 expansion • **Target 3:** 2.618 runner.

QUANTITATIVE SHORT SCREENING STRATEGY

VERTICAL RUN-UP BLOW-OFFS, WYCKOFF UTAD & REALISTIC BOUNDED BRACKETS

Direction: SHORT Setups
Horizon: Multi-Day Swing (1–5 Days)
Target Bounding: Max 1.75R / Max 14% Drop

1. THE SHORT ENGINE PHILOSOPHY: NEVER SHORT SUPPORT FLOORS

A common flaw in algorithmic screeners is shorting assets that have already suffered severe multi-week downtrends and are grinding into support floors (e.g. TRB, AXS, HBAR at range lows). These setups offer poor risk/reward and suffer vicious short-squeeze spikes. The Atlas Short engine enforces that **shorts are sold HIGH**—specifically targeting overextended assets in the **upper 55% to 100% of their 30-day range following a confirmed vertical run-up** where late retail buyers are trapped.

2. THE 5 MANDATORY QUANTITATIVE SHORT GATES

Gate Name	Mathematical Formulation	Institutional Function
Gate 1: Mandatory Macro Run-Up	`Range_Height >= 16%` or `Ret_7d >= 10%` or `Ret_14d >= 14%` or `(Close - EMA50_4H) >= +8%`	**Distribution requires prior markup.** Eliminates flat, lifeless, range-bound tokens that never experienced speculative vertical expansion.
Gate 2: Strict Upper-Range Positioning	`Pos_in_Range = (Close - Low_30d) / (High_30d - Low_30d)` **Mandate:** `Pos_in_Range >= 0.55`	Guarantees shorts are initiated strictly in the upper half of the range near distribution ceilings. Rejects bottom-chopping tokens.
Gate 3: Anti-Chasing & Oversold Filter	`Ret_24h >= -18% & Ret_7d >= -35%` `RSI_1H >= 28 & RSI_4H >= 38`	Prevents shorting into deeply exhausted bottoms or selling into capitulation wicks where mean-reversion bounces occur.
Gate 4: Multi-Timeframe Bearish Divergence	`RSI_4H < Prev_RSI_4H` on Higher High 4H MACD Histogram Bearish Divergence	Detects institutional exhaustion: price prints a nominal higher high, but momentum indicators decisively print lower peaks.
Gate 5: Trapped Longs & OI Crowding	`Funding_Rate > 0` (Longs paying) Large positive 7d OI (+15% to +200%)	Identifies overcrowded retail longs paying funding fees into smart-money distribution wicks, providing fuel for cascading liquidations.

3. SHORT SETUP ARCHETYPES

`UTAD_DIVERGENCE_SHORT` **Phase C**

Wyckoff Upthrust After Distribution: Price sweeps the 30-day high with buying climax wicks, followed by confirmed 4H/Daily bearish divergence and immediate rejection back inside the range.

`DISTRIBUTION_SHORT` **Macro Rollover**

Parabolic Markup Breakdown: Multi-week runners (>75% to 240% expansion) that coil below the 4H EMA50 and 30-day POC, displaying heavy seller volume dominance and structural momentum decay.

`EXHAUSTION_REVERSAL_SHORT` **Climax**

Multi-Timeframe Overbought Blow-Off: Assets where Weekly and Daily RSI exceed 75–84 with massive open interest surges (+50% to +216% 7d OI), ready for an immediate mean-reversion cascade.

4. REALISTIC SHORT EXECUTION BRACKETS (ELIMINATING ANCIENT WICK OVERSHOOT)

In volatile crypto pairs, ancient multi-month swing wicks often generated absurd downside targets (-55% to -80% below current price). Atlas v2.6 enforces realistic geometric bounding:

```
BOUNDED SHORT TARGET: Target_1 = max(Structural_Support, Entry_Price - 1.75 * Risk_Dist, Entry_Price * 0.86)
```

- **Target 1 Capped at 1.75R / Max 14% Drop:** Ensures attainable swing targets rather than distant historic support artifacts.
- **Retest Entry (Sell-Limit):** Set at the 4H EMA50 / POC retest shelf (`POC_Price - 0.15 * ATR`).
- **Breakdown Trigger (Stop-Market):** Set below the local consolidation floor (`Local_Low - 0.05 * ATR`).
- **Invalidation Stop:** Placed structurally above the local distribution swing high / VAH ceiling (`Resistance + 0.35 * ATR`).

SCORING ENGINE & INSTITUTIONAL QUALITY GATES

MULTI-FACTOR SCORING, STRICT GRADE A/B FLOOR & CURATED TOP 16 SIZING

Model: Non-Linear Factor Weighting
Floor: Grade B (Score >= 68.0, Conviction >= 65.0)
Desk Cap: 16 Curated Institutional Picks

1. MULTI-FACTOR REGRESSION FORMULA

Every candidate passing the hard structural gates is evaluated through a non-linear composite scoring engine (0.0 to 100.0) across five orthogonal quantitative dimensions:

```
SCORE = 0.25 * Trend_Score + 0.25 * Momentum_Score + 0.25 * Derivatives_Score + 0.15 * Alpha_Score + 0.10 * Volume_Score
```

Factor Dimension	Weight	Underlying Quantitative Metric Inputs
Trend Architecture	25%	4H EMA20 vs EMA50 alignment, ADX trend strength (≥25), DMI directionality, Minervini stage positioning.
Momentum & Squeeze	25%	Bollinger Bandwidth (<10%), BB nested inside KC, MACD histogram velocity, normalized price momentum.
Derivatives Positioning	25%	7-day & 1-hour Open Interest velocity, contrarian funding rate arbitrage, taker buy/sell ratio, order book depth imbalance.
BTC-Residual Alpha	15%	Residual alpha vs Bitcoin over 6h, 24h, and 7d horizons; beta adjustment to isolate idiosyncratic token momentum.
Volume Microstructure	10%	Relative volume surge (RVOL ≥ 1.25x), volume contraction ratio (down-bar dry-up), 24h quote volume depth.

2. COMPOSITE CONVICTION & STRICT QUALITY GATING

To prevent the screener from degrading with marginal, below-threshold setups, Atlas enforces strict **Institutional Quality Gates**:

```
CONVICTION = [ 0.40 * Estimated_Win_Rate_Pct + 0.60 * Score ] * Consensus_Multiplier
```

- **Hard Veto on "Below Threshold":** Zero candidates with substandard technical structures are permitted into the watchlist.
- **Institutional Floor:** Composite Score ≥ 68.0, Conviction ≥ 65.0, $R:R \ge 2.0$, and Expected Value $E[V] \ge +0.20R$.
- **Consensus Consensus Multiplier:** 5-System Consensus earns a 1.08x bonus; 4-System earns 1.04x; ≤ 2 systems penalized.
- **Curated Desk Sizing:** Watchlist strictly capped at **16 high-conviction setups** (up to 12 Longs, up to 6 Shorts), deduplicated by base asset to eliminate multi-pair dilution.

3. THE 5-POINT INSTITUTIONAL CHECKLIST

Every candidate must undergo a deterministic 5-point verification audit displayed as badges on the trader dashboard:

1. VALUE ACCEPTANCE / BREAKDOWN
Verified trading above 30d POC (Long) or rejected below 30d POC / 4H EMA50 (Short).

2. VOLATILITY COMPRESSION / UTAD
Verified BB inside KC squeeze (Long) or confirmed 4H/Daily bearish divergence (Short).

3. MOMENTUM & TREND ALIGNMENT
4H EMA20/50 alignment and MACD histogram acceleration in trade direction.

4. VOLUME CONTRACTION / EXHAUSTION
Volume drying up on pullbacks (VCR ≤ 1.0x) or volume surge on breakout trigger.

5. SMART MONEY ORDER FLOW & DERIVATIVES POSITIONING
Non-negative 7d OI on long breakouts; trapped overleveraged retail longs (+15% to +200% OI, positive funding) on short setups.

MACRO INTELLIGENCE & OPERATIONAL TOPOLOGY

MARKET OVEREXTENSION ARCHITECTURE, ALIGNED 4H CRON & SYSTEM RESILIENCE

Service: Macro Liquidity & Radar Port
Cadence: 4-Hour UTC Candle-Aligned Cron
Performance: 21.6s Full Sweep (86% CPU Reduction)

1. MACRO OVEREXTENSION ARCHITECTURE (MOEA)

The platform incorporates a Macro Market Intelligence engine to determine overall market regimes and compute the probability of a market-wide sell-off or cool-off. The **Composite Overextension Index (COI: 0–100)** unifies four distinct market pillars:

```
COI = 0.35 * Benchmark_Excursion + 0.30 * Leverage_Crowding + 0.20 * Market_Breadth + 0.15 * Stablecoin_Liquidity
```

1. Benchmark Excursion (35%)
BTC & ETH distance from Daily EMA50 and Daily Bollinger Upper Band, plus 14-day RSI excursion.

2. Leverage & Funding Crowding (30%)
Annualized perpetual funding rates across the top 20 perpetuals. Rates > 25% APR indicate extreme long vulnerability.

3. Market Breadth & Euphoria (20%)
Percentage of altcoins trading above their 20-day moving average combined with Alternative.me Fear & Greed index.

4. Stablecoin Fuel vs Leverage (15%)
7-day net issuance flow of USDT/USDC via DeFiLlama. Divergence between price gains and stablecoin outflows signals synthetic leverage.

2. AUTOMATED 4-HOUR CRON SERVICE ALIGNED WITH CANDLE CLOSES

The screener background daemon executes every 4 hours, synchronized precisely with global Binance USD-M candle closes and funding intervals:

- **Aligned UTC Intervals: 00:00 · 04:00 · 08:00 · 12:00 · 16:00 · 20:00 UTC** (+30 second candle close grace).
- **Operational Rationale:** 4H candle closures represent the primary technical decision boundary for institutional algorithmic strategies.
- **Zero-Lag Cache Persistence:** Scan results are persisted to `data/screener_latest.json` and served in <10ms to trading dashboards.

3. PRODUCTION INFRASTRUCTURE & RESILIENCE MATRIX

Infrastructure Layer	Technology Implementation	Fault Isolation Guarantee
Vectorized Indicators	Inlined float triangular VPVR with precomputed inverse spans.	Per-token VPVR drops from 474ms to 66ms (86% CPU speedup), avoiding event-loop blocking.
2-Stage Derivatives	Stage 1 structural filtering; Stage 2 isolated derivatives snapshot.	Reduces external Binance HTTP calls by >85%; transient API errors handled via exception masking.
In-Memory Resampling	1D/1W candles generated in-memory directly from 4H bars.	Eliminates 360 redundant HTTP requests per scan cycle.
Deployment Topology	Docker bind-mount (`./src:/app/src:ro`) with live uvicorn reload.	Code updates take effect immediately on container restart without slow wheel rebuilds.